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Rh-Catalyzed cyclization of 2,3-allenoic acids in the presence of 2,3-allenols†‡

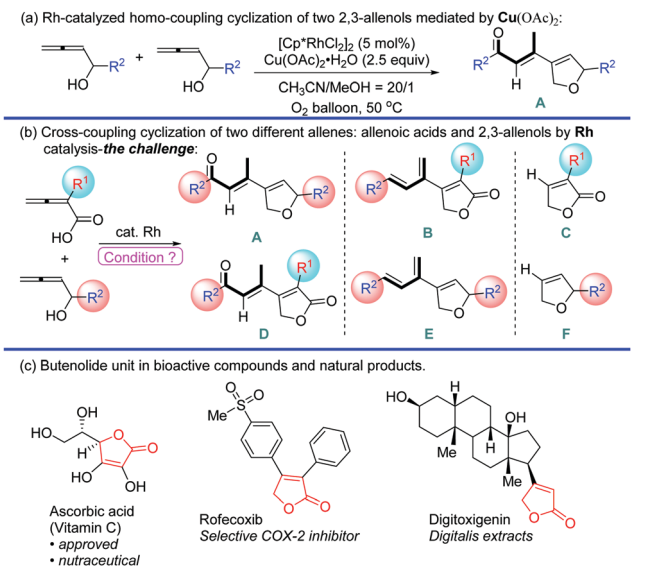
Junjie Fan,[a] Chunling Fu,[a] Xiaoyan Wu *[ab] and Shengming Ma *[ab]

Herein we report a [Cp*RhCl$_2$]$_2$-catalyzed coupling cyclization of two different classes of allenes with 2,3-allenoic acids affording 2(5*H*)-furanone skeletons of products and 2,3-allenols forming a conjugated (*E*)-enal or enone functionality to the β-position of the 2(5*H*)-furanones. These products are important building blocks for the syntheses of potentially bioactive compounds. The reaction proceeded *via* the nucleometalation, insertion, and stereodefined 1,4-*H* delivery carried by rhodium.

Over the past few decades, the chemistry of allenes has been receiving more and more attention.[1] Now it has become a common approach to construct different types of cyclic structures with synthetic or bio-potentials *via* the cyclizations of allenes with a nucleophilic functionality.[2–4] Recently, Rh-catalyzed cyclization of two molecules of the same 2,3-allenols was reported to afford (2,5-dihydrofuran-3-yl)but-2-(*E*)-*en*-1-one derivatives **A** (Scheme 1a).[5] Here we envisioned such a bimolecular allene cyclization reaction of two different types of allenes. In the first place, we wished to attempt the cyclization of 2,3-allenoic acids in the presence of 2,3-allenols. There are intricate challenges: (1) as expected, both homo-coupling cyclization 2,5-dihydrofuran products **A** and **E** or the cyclizative cross-coupling furanone products **B**[4a] and **D** may be formed (Scheme 1b); (2) in addition, the cycloisomerization products **C**[4b] and **F**[4c] may also be formed. We wish to report here a highly selective protocol of the cyclizative cross-coupling of allenoic acids in the presence of 2,3-allenols to construct a new type of furanone derivatives, which are common units existing in many bioactive compounds and natural products, for instance, ascorbic acid (vitamin C),[6] the selective COX-2 inhibitor: Rofecoxib,[7] digitalis extracts: digitoxigenin[8] *etc.* (Scheme 1c).

We began our studies on the reaction of 2,3-allenoic acid **1a** and 2,3-allenol **2a** under the conditions for the homo-cyclizative coupling products of allenols (Scheme 2).[5] The reaction did afford the homo-coupling product (*E*)-**3a** in 9% NMR yield derived from two molecules of 2,3-allenol **2a**. In addition, the desired cross-over cyclizative coupling product (*E*)-**4aa** was formed in 29% NMR yield. The **B**-type diene product **5aa** and cycloisomerization product **6a** and **7a** were not detected.

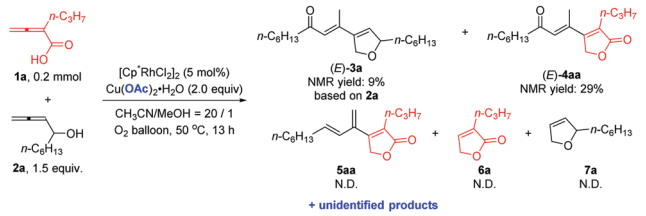
In order to optimize the selectivity for the formation of a cross-coupling product (*E*)-**4aa**, $CuCl_2{\cdot}2H_2O$, $Cu(NO_3)_2$ and CuO were applied as the oxidants to replace $Cu(OAc)_2{\cdot}H_2O$ under a nitrogen atmosphere (Table 1, entries 1–3): with $CuCl_2{\cdot}2H_2O$, the cycloisomerization product **6a** of **1a** was formed in 2% NMR yield (entry 1),



Scheme 1 Rh-catalyzed reactions of two allenes and selected examples of butenolides.

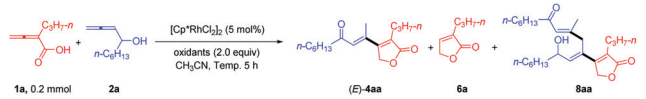
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Scheme 2 Rh-catalyzed cross-coupling of allenoic acid **1a** with 2,3-allenol **2a**.

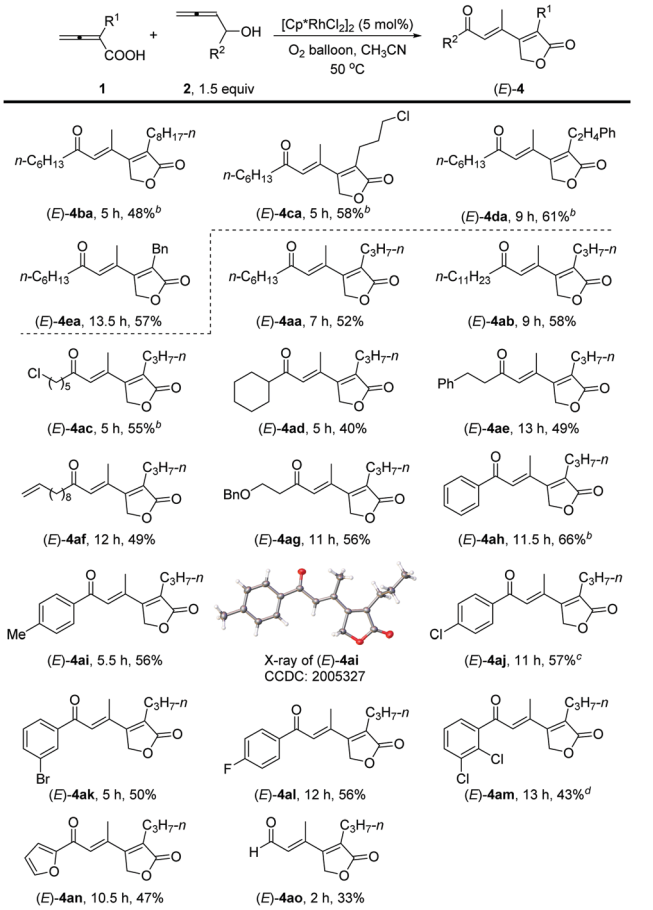
Table 1 Optimization studies for cyclizative cross-coupling of allenoic acids **1a** with 2,3-allenols **2a** getting compound (*E*)-**4aa**[a]



				NMR Yield[b] (%)		
Entry	Oxidants	Temp. (°C)	**1a**/**2a**	(*E*)-**4aa**	**6a**	**8aa**
1[c,d]	$CuCl_2·2H_2O$	50	1.5/1	0	2	0
2[c]	$Cu(NO_3)_2$	50	1.5/1	21	0	0
3[c]	CuO	50	1.5/1	20	0	0
4	CuO + O_2	50	1.5/1	28	2	3
5	O_2	50	1.5/1	39	12	4
6[e]	O_2	50	1/1.3	52	2	7
7[f]	O_2	50	1/1.5	61	0	7
8[f]	O_2	50	1/1.8	51	0	6
9[f,g]	O_2	30	1/1.5	52	0	6
10[f]	O_2	60	1/1.5	59	2	8
11[c]	—	50	1/1.5	0	0	0

[a] The reaction was conducted on 0.2 mmol scales, 2,3-allenoic acid **1a**, 2,3-allenol **2a**, $[Cp^*RhCl_2]_2$ (5 mol%) and oxidants (2.0 equiv.) in CH_3CN (0.54 mL) for 5 h unless otherwise noted. [b] Determined by ^1H NMR analysis of the crude products using mesitylene as the internal standard. [c] Under N_2 atmosphere. [d] 69% of 4-chloro-3-propyl-2(5*H*)-furanone was detected. [e] 1% of (*E*)-**3a** was detected. [f] 2% of (*E*)-**3a** was detected. [g] 12 h, 4% recovery of **1a**.

Table 2 Reaction scope[a]

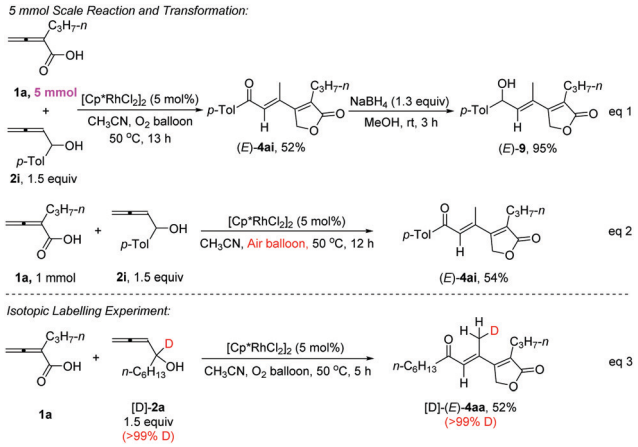


[a] The reaction was conducted with 2,3-allenoic acid **1** (1.0 mmol), 2,3-allenol **2** (1.5 equiv.) and $[Cp^*RhCl_2]_2$ (5 mol%) in CH_3CN (2.7 mL) at 50 °C under an O_2 atmosphere (balloon) unless otherwise noted. 2–3% of (*E*)-**3** and 5–8% of **8** were detected in the reaction of **1** with **2** except the reaction of **1a** with **2o**. [b] 0.5 mmol scale. [c] *E*/*Z* = 94/6. [d] *E*/*Z* = 97/3.

meanwhile, 69% NMR yield of 4-chloro-3-propyl-2(5*H*)-furanone was detected. The C–Cl bond should be formed *via* reductive elimination of the intermediate with a C–M–Cl unit.[9] With $Cu(NO_3)_2$ and CuO, target product (*E*)-**4aa** was obtained in 21% and 20% yields (entries 2 and 3). Using oxygen instead of nitrogen could improve the yield. The reaction of **1a** in the presence of **2a** afforded (*E*)-**4aa** in 28% NMR yield with the generation of two by-products: **6a** and the trimolecular cascade product **8aa** (entry 4). **8aa** was generated from one molecule of **1a** and two molecules of **2a**. When using oxygen as the only oxidant, the NMR yield of (*E*)-**4aa** was improved to 39% with a substantial increase of **6a** to 12% NMR yield (entry 5). By adjusting the ratio of **1a**/**2a** from 1.5/1 to 1/1.8 (entries 5–8), the reaction afforded (*E*)-**4aa** in the highest yield of 61% with the ratio of 1/1.5, together with **8aa** in 7% yield; no **6a** was formed (entry 7). The reaction at 30 °C was sluggish and the yield dropped to 52% (entry 9). The reaction at 60 °C afforded (*E*)-**4aa** in a slightly lower yield (entry 10). Control experiments under a nitrogen atmosphere indicted that oxygen is essential for this transformation (entry 11). Thus, we defined the conditions of entry 7 for highly selective formation of (*E*)-**4**.

The reaction scope under optimal conditions was then investigated. Furanone derivatives containing the synthetically versatile α,β-enals and enones (*E*)-**4** were synthesized highly *E*-selectively (Table 2). R^1 groups of 2,3-allenoic acids may be *n*-octyl, 3-chloropropyl, phenethyl, and benzyl, reacting with allenol **2a** to afford (*E*)-**4ba**, (*E*)-**4ca**, (*E*)-**4da**, and (*E*)-**4ea** in 48–61% yields. When R^2 are 1°-alkyl or 2°-alkyl groups, the reaction with 2,3-allenoic acid **1a** gave moderate yields of (*E*)-**4aa**, (*E*)-**4ab**, (*E*)-**4ad**, and (*E*)-**4ae**. Functionalized allenols with R^2 being 5-chloropentyl, terminal C=C bond, and benzyl protected hydroxyl showed decent reactivity to afford (*E*)-**4ac**, (*E*)-**4af**, and (*E*)-**4ag** in 49–56% yields. Phenyl and halogen atom-substituted (F, Cl, Br) on the *para*-, *ortho*- or *meta*-position of the benzene ring could all be tolerated to afford (*E*)-**4ah**–(*E*)-**4am** in 43–66% yields. We confirmed the structure of the products by single crystal of **4ai**. 2-Furyl-substituted allenol **2n** also could react with allenoic acid **1a** to produce (*E*)-**4an** in 47% yield. In addition, α,β-unsaturated enal (*E*)-**4ao** may be prepared from buta-2,3-dien-1-ol **2o** and **1a** in 33% yield.

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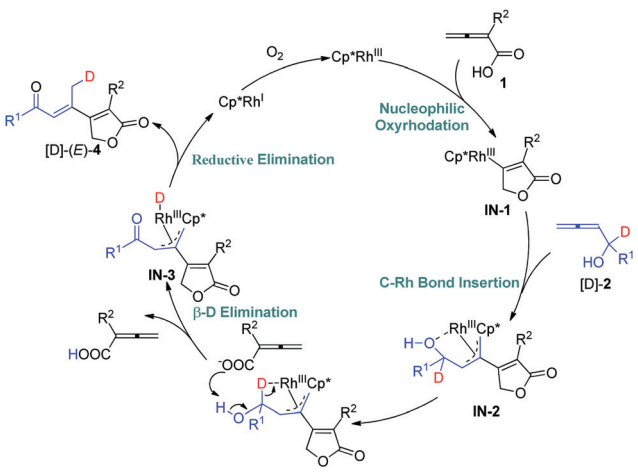


Scheme 3 5 mmol scale reaction, transformation and isotopic labeling experiment.

To further explore the synthetic utility of this methodology, the reactions were conducted on 5 mmol scale under standard conditions to afford (*E*)-**4ai** in 52% yield. (*E*)-**4ai** could be reduced to an *E*-selective allyl alcohol (*E*)-**9** by treatment with sodium borohydride (Scheme 3, eqn (1)). When the oxygen atmosphere was replaced with an air atmosphere, the reaction worked similarly, affording (*E*)-**4ai** in 54% yield (Scheme 3, eqn (2)).

To clarify the mechanism, an isotopic labelling experiment was conducted (Scheme 3, eqn (3)): the reaction of allenoic acid **1a** and deuterated allenol [D]-**2a** (>99% D) under standard conditions afforded [D]-(*E*)-**4aa** with the methyl connected to the sp^2 carbon atom being mono-deuterated, indicating the possibility of a 1,4-H delivery pathway.[10]

Based on the experimental results and our previous studies, a mechanism was proposed as shown in Scheme 4. At first, the allenoic acid **1** went through a nucleophilic cyclic oxyrhodation to afford 2-oxo-2,5-dihydrofuran-4-yl rhodium intermediate **IN-1**. Subsequently the allene unit in deuterated allenol [D]-**2** underwent *syn*-insertion into the carbon–rhodium bond and



Scheme 4 Proposed mechanism for cyclizative cross-coupling.

the OH group coordinated with Rh to generate the stereodefined allylic rhodium intermediate **IN-2**. The deuterated allylic rhodium hydride **IN-3** was formed by deprotonation of the hydroxyl group mediated by an allenoate anion[10] and β-D elimination. **IN-3** would yield [D]-(*E*)-**4** and Rh(I) species *via* reductive elimination. The Rh(I) species would be oxidized to Rh(III) by O_2 to finish the catalytic cycle.[11]

In conclusion, the Rh(III)-catalyzed highly chemoselective cyclizative cross-coupling of 2,3-allenoic acids with 2,3-allenols affording 4-(4′-oxo-2′-*en*-2′-yl)-2(*5H*)-furanone derivatives was established. The reaction proceeded *via* cyclic nucleometalation, insertion, and stereodefined 1,4-*H* delivery as unveiled by mechanistic studies. Further studies in this area are currently ongoing in this laboratory.

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Conflicts of interest

There are no conflicts to declare.

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